

Seongmin Hong

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EDUCATION

Mar. 2019 - Present	Kyung Hee University Bachelor of Electronic Engineering, Computer Engineering GPA 4.14/4.5	Yongin-si, Korea
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SKILLS & INTERESTS

- **Programming:** C++, Python, ROS1, ROS2, Git, PyTorch
- **Research Interest:** Autonomous Driving, Mobile Robotics, Navigation, Deep Learning

RESEARCH EXPERIENCE

Jul. 2025 - Aug. 2025	Autonomous Driving Intelligence Lab <i>Research Intern</i> - Auto-labeling tool that extracts lanes/trajectories/actors/work-zones from surround-view + LiDAR + HD maps, generating VQA fine-tuning <i>JSONL</i>. - Vision-fine-tuned InternVL 3.0 and GPT-4o/5 (API) to produce scene-understanding and action-decision outputs for E2E autonomous driving	ETRI, Korea
Mar. 2023 - May. 2024	Immersion <i>Autonomous Driving Developer</i> - Led Perception & Navigation; built an end-to-end outdoor autonomy stack from hardware design and sensing to path planning and control. - Designed and integrated a prediction-based avoidance module, improving dynamic obstacle handling in the existing autonomy stack.	Seoul, Korea
Jul. 2021 - Aug. 2022	Human-Robot Interaction Lab <i>Undergraduate Research Assistant</i>	Kyung-Hee University, Korea

PUBLICATIONS

Apr. 2025	Development of Autonomous Pick-and-Place Robot System with Multi-Floor Navigation Journal of Korea Robotics Society (Under revision) Seongmin Hong , Yujin Jung, Yunjeong Choi, Seju Jang, Donghan Kim
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AWARDS

Mar.2025	Academic Excellence Award (Top Rank in Department) • Kyung-Hee University, College of Electronics and Information
Jan.2025	Grand Prize, Capstone Design Competition • LINC 3.0 Project Group, Kyung-Hee University
Jan.2025	Excellence Award, Graduation Project • Kyung-Hee University, College of Electronics and Information
Jul.2024	Encouragement Award, Autonomous Aviation Mini-drone Competition • Korean Institute of Electrical Engineers

Jul.2022

2nd Prize, Autonomous Aviation Mini-drone Competition

- Korean Institute of Electrical Engineers

PROJECTS

1. Enhancing Diffusion-Based End-to-End Autonomous Driving with VLM

Mar.2025 – Jun.2025

 - Improved trajectory diversity and performance with VLM-guided dynamic anchor sampling; switched to a lightweight YOLO sampler for real-time with comparable results.
 - Built a VLM-based Trajectory Scorer that re-ranks candidates by centerline alignment and collision-avoidance metrics, improving final path selection.
2. Autonomous Pick-and-Place Robot System with Multi-Floor Navigation (Husky + UR3)

Jul.2024 – Mar.2025

 - Developed an autonomous multi-floor pick-and-place robot system using a Husky-UR3
 - Applied hybrid maps, AI vision, AR-marker, etc., to ensure safe navigation and accurate button press
3. Outdoor Advertising Robot (Immersion Project)

Nov.2023 – Mar.2024

 - Enhanced dynamic obstacle avoidance by predicting obstacle motion using ByteTrack + Kalman filter
 - Developed a GUI-linked behavior tree for flexible and user-driven navigation control
4. Mando Autonomous Driving Challenge

Oct.2023 – Nov.2023

 - Combined dual camera images using real-time SURF-based stitching for wide-lane coverage
 - Unified multi-source datasets into YOLOv8 format and trained a custom model for signal and pedestrian detection
5. Woowa Bros. Robot Delivery Challenge

Sep.2023 – Oct.2023

 - Stabilized GPS-denied localization by fusing 3D LiDAR-based pose and IMU on a prebuilt 3D map
 - Calculated pedestrian speed and direction from YOLOv5 and LiDAR data, integrating collision risk into DWA for safe local path planning
6. Lane-Keeping System in CARLA Simulator

May.2023 – Jun.2023

 - Developed a robust lane-keeping system using YOLOv2, BEV, and Kalman filtering
 - Applied quadratic curve fitting and Stanley controller to compute accurate steering angles

LANGUAGES

English

Level 2 – High Intermediate (NEW TEPS: 348)

Korean

Native